Cryptography and Web Security

GTU Diploma Engineering – Unit Notes

Note: These notes follow the uploaded GTU syllabus and use simple, beginner-friendly, exam-oriented language.

UNIT 1 – Introduction of Information & Network Security

1.1 Computer Security

Definition: Computer security means protecting a computer system and its resources from unauthorized access, damage, misuse, or theft.

3 Uses:

- Protects files, programs, and system resources.
- Prevents unauthorized users from accessing the computer.
- Helps keep computer systems safe during daily use.

Applications & Real-Life Example: Applications: Used in personal computers, offices, banks, schools, and organizations. Real-life example: A computer login password prevents an unauthorized person from opening a student's files.

1.1 Cyber Security

Definition: Cyber security means protecting computers, networks, devices, and digital information from attacks or unauthorized activities.

3 Uses:

- Protects digital systems from cyber attacks.
- Helps secure networks and connected devices.
- Protects important digital information.

Applications & Real-Life Example: Applications: Used in websites, mobile applications, companies, banks, and government systems. Real-life example: A bank uses cyber-security controls to protect online banking accounts from attackers.

1.1 Information Security

Definition: Information security means protecting information from unauthorized access, change, disclosure, or loss. It focuses on keeping information secure.

3 Uses:

- Protects important information.
- Helps maintain confidentiality of data.
- Helps protect data from unauthorized changes.

Applications & Real-Life Example: Applications: Used in hospitals, banks, colleges, companies, and government offices. Real-life example: A hospital protects patient records so that only authorized staff can access them.

1.1 Network Security

Definition: Network security means protecting a computer network and the data moving through it from unauthorized access and attacks.

3 Uses:

- Protects network communication.

- Controls unauthorized network access.
- Helps protect data transmitted over a network.

Applications & Real-Life Example: Applications: Used in Wi-Fi networks, company networks, data centers, and internet services. Real-life example: A Wi-Fi password prevents unauthorized devices from joining a private network.

1.2.1 Confidentiality

Definition: Confidentiality means that information should be available only to authorized persons or systems. It prevents unauthorized disclosure of information.

3 Uses:

- Keeps private information secret.
- Allows access only to authorized users.
- Protects sensitive data from unauthorized disclosure.

Applications & Real-Life Example: Applications: Used for passwords, banking information, personal records, and confidential business data. Real-life example: A student's examination result is visible only to authorized users through a secure login.

1.2.2 Integrity

Definition: Integrity means keeping information correct, complete, and unchanged unless an authorized person makes the change.

3 Uses:

- Keeps data accurate.
- Prevents unauthorized modification.
- Helps maintain complete information.

Applications & Real-Life Example: Applications: Used in banking, databases, examination systems, and online transactions. Real-life example: A bank must ensure that a transferred amount is not changed during processing.

1.2.3 Availability

Definition: Availability means that authorized users can access information and services whenever they are needed.

3 Uses:

- Keeps services accessible.
- Supports continuous system operation.
- Allows authorized users to obtain required data.

Applications & Real-Life Example: Applications: Used in websites, banking systems, servers, and online services. Real-life example: An online banking website should be available when customers need to check their accounts.

1.2.4 Authenticity

Definition: Authenticity means verifying that a user, device, message, or data source is genuine and really belongs to the claimed identity.

3 Uses:

- Verifies identity.
- Checks whether data or users are genuine.
- Reduces unauthorized access.

Applications & Real-Life Example: Applications: Used in login systems, digital communication, banking, and websites. Real-life example: A website asks for a username and password to verify the identity of a user.

1.2.5 Accountability

Definition: Accountability means that actions performed by a user or system can be traced to the responsible person or entity.

3 Uses:

- Records user activities.
- Helps trace actions to users.
- Supports monitoring of system use.

Applications & Real-Life Example: Applications: Used in organizations, banking systems, servers, and databases. Real-life example: A system log records which employee changed a particular file.

1.3 OSI Security Architecture

Definition: OSI Security Architecture is a framework that provides a structured way to understand security attacks, security services, and security mechanisms in network communication.

3 Uses:

- Classifies security attacks.
- Describes security services.
- Describes security mechanisms used for protection.

Applications & Real-Life Example: Applications: Used for understanding and designing secure network communication. Real-life example: Network engineers use security concepts to decide how authentication and encryption should protect communication.

UNIT 2 – Number Theory in Cryptography

2.1 Divisibility and Division Algorithm

Definition: Divisibility describes whether one integer can divide another integer without a remainder. The division algorithm expresses a number using a quotient and a remainder.

3 Uses:

- Helps perform integer calculations.
- Used to find quotient and remainder.
- Forms a basis for GCD calculations.

Applications & Real-Life Example: Applications: Used in cryptographic calculations and number theory. Real-life example: 17 divided by 5 gives quotient 3 and remainder 2, so 17 = 5 × 3 + 2.

2.2 Modulo Operator

Definition: The modulo operator gives the remainder after one integer is divided by another integer. It is written using the symbol %.

3 Uses:

- Finds remainders.
- Used in modular arithmetic.
- Used in cryptographic calculations.

Applications & Real-Life Example: Applications: Used in encryption algorithms, number theory, and cyclic calculations. Real-life example: 17 % 5 = 2, because the remainder after dividing 17 by 5 is 2.

2.3 Set of Residues Zn

Definition: The set of residues Zn contains the possible remainder values from 0 to n−1 when numbers are considered modulo n.

3 Uses:

- Represents numbers under a modulus.
- Supports modular calculations.
- Used in cryptographic algorithms.

Applications & Real-Life Example: Applications: Used in modular arithmetic and cryptography. Real-life example: For Z5, the residue set is {0,1,2,3,4}.

2.4 GCD and Multiplicative Inverse

Definition: The Greatest Common Divisor (GCD) is the largest positive integer that divides two numbers. A multiplicative inverse of a number modulo n is a value that gives 1 when multiplied with the number modulo n.

3 Uses:

- Used in number-theory calculations.
- Helps check whether an inverse exists.
- Used in cryptographic key calculations.

Applications & Real-Life Example: Applications: Used in RSA and other cryptographic calculations. Real-life example: The Euclidean algorithm can be used to calculate the GCD of 48 and 18 as 6.

UNIT 3 – Classical Encryption Techniques

3.1 Symmetric-key Encryption

Definition: Symmetric-key encryption is a method in which the same secret key is used for encryption and decryption of data.

3 Uses:

- Uses a shared secret key.
- Encrypts plaintext into ciphertext.
- Uses the key again to decrypt ciphertext.

Applications & Real-Life Example: Applications: Used for protecting stored data and secure communication. Real-life example: A file can be encrypted with a secret key and later opened using the same key.

3.1.3 Brute-Force Attack

Definition: A brute-force attack tries possible keys one by one until the correct key is found and the encrypted message can be understood.

3 Uses:

- Tries many possible keys.
- Can be used to test weak keys.
- Works by systematic key searching.

Applications & Real-Life Example: Applications: Used in cryptanalysis and security testing. Real-life example: If an encryption key has a very small number of possibilities, an attacker may try every possible key.

3.2 Substitution Cipher

Definition: A substitution cipher replaces each plaintext character with another character according to a rule or key.

3 Uses:

- Replaces plaintext characters.
- Uses a substitution rule or key.
- Produces ciphertext from plaintext.

Applications & Real-Life Example: Applications: Used for learning classical cryptography and understanding substitution-based encryption. Real-life example: In an additive cipher, each letter can be shifted by a fixed number of positions.

3.3 Transposition Cipher

Definition: A transposition cipher changes the order or position of plaintext characters without replacing the characters themselves.

3 Uses:

- Rearranges characters.
- Keeps the original characters.
- Uses a particular arrangement or key.

Applications & Real-Life Example: Applications: Used in classical cryptography and for learning encryption concepts. Real-life example: The letters of a message can be written in rows and then read column by column to create ciphertext.

3.3.3 Stream and Block Ciphers

Definition: A stream cipher encrypts data one bit or byte at a time, while a block cipher encrypts data in fixed-size blocks.

3 Uses:

- Stream ciphers process data continuously.
- Block ciphers process fixed-size blocks.
- Both convert plaintext into ciphertext.

Applications & Real-Life Example: Applications: Used in secure data communication and encryption systems. Real-life example: A block cipher can process a long file by dividing it into fixed-size blocks.

3.4 Asymmetric-key Cryptosystem

Definition: An asymmetric-key cryptosystem uses two related keys: a public key and a private key. The keys perform different roles in encryption and decryption.

3 Uses:

- Uses a public key.
- Uses a private key.
- Supports secure communication without sharing one secret key.

Applications & Real-Life Example: Applications: Used in secure communication, authentication, and digital security. Real-life example: A person can use a website's public key as part of establishing secure communication.

3.4.3 RSA Algorithm

Definition: RSA is an asymmetric encryption algorithm that uses a public key and a private key. It is based on mathematical operations involving large numbers and their factors.

3 Uses:

- Uses two related keys.
- Supports public-key encryption.
- Uses mathematical calculations for key generation and encryption.

Applications & Real-Life Example: Applications: Used in public-key cryptography and secure communication. Real-life example: RSA can be used to protect a secret value while communicating between two systems.

UNIT 4 – Web Security

4.1 Web Security

Definition: Web security means protecting websites, web applications, users, and web communication from threats and unauthorized activities.

3 Uses:

- Protects web applications.
- Protects data exchanged through websites.
- Helps reduce web-based security threats.

Applications & Real-Life Example: Applications: Used in online banking, shopping websites, social media, and web applications. Real-life example: A shopping website uses secure communication to protect customer information during online transactions.

4.2 SSL and TLS

Definition: SSL and TLS are security protocols designed to protect communication between a client and a server. TLS is the newer protocol family used for secure communication.

3 Uses:

- Protects data during communication.
- Helps provide secure connections.
- Supports secure client-server communication.

Applications & Real-Life Example: Applications: Used by websites and online services for secure communication. Real-life example: A browser can establish a secure connection with a website using TLS.

4.3 HTTPS

Definition: HTTPS stands for Hypertext Transfer Protocol Secure. It uses HTTP with security protection to provide safer communication between a web browser and a web server.

3 Uses:

- Protects web traffic.
- Uses secure communication protocols.
- Helps protect information sent between browser and server.

Applications & Real-Life Example: Applications: Used by banking, shopping, login, and other websites. Real-life example: When a user logs into an online banking website, HTTPS helps protect the login information during transmission.

4.4 Secure Electronic Transactions

Definition: Secure Electronic Transaction (SET) is a security concept and protocol framework designed to protect electronic payment transactions.

3 Uses:

- Protects electronic payment information.
- Supports secure transaction processing.
- Uses cryptographic security concepts.

Applications & Real-Life Example: Applications: Used as a concept for understanding secure electronic payment systems. Real-life example: An electronic payment system needs to protect payment information while a customer makes an online purchase.

UNIT 5 – System Security

5.1 Intrusion Detection

Definition: Intrusion detection is the process of identifying unauthorized or suspicious activities in a computer system or network.

3 Uses:

- Detects suspicious activities.
- Monitors system or network activity.
- Helps identify unauthorized access.

Applications & Real-Life Example: Applications: Used in organizations, servers, networks, and security systems. Real-life example: An intrusion detection system can alert an administrator when unusual network activity is detected.

5.1.2 Intruders

Definition: An intruder is a person or entity that attempts to access a computer system or network without proper authorization. The syllabus classifies intruders as Masquerador, Misfeasor, and Clandestine user.

3 Uses:

- Attempts unauthorized access.
- May misuse system resources.
- Can be classified according to the type of unauthorized activity.

Applications & Real-Life Example: Applications: Used in security analysis and intrusion detection. Real-life example: A person using another user's stolen login credentials can act as a masquerador.

5.1.3 Intrusion Detection System

Definition: An Intrusion Detection System (IDS) is a security system that monitors activities and identifies possible unauthorized or suspicious behavior.

3 Uses:

- Monitors activities.
- Detects possible intrusions.
- Can use misuse or anomaly detection approaches.

Applications & Real-Life Example: Applications: Used in computer networks, servers, and organizations. Real-life example: An IDS can alert a network administrator when traffic matches a known attack pattern.

5.2 Firewall

Definition: A firewall is a security mechanism that controls network traffic between trusted and untrusted networks according to defined security rules.

3 Uses:

- Controls network traffic.
- Helps block unauthorized access.
- Monitors communication between networks.

Applications & Real-Life Example: Applications: Used in computers, offices, servers, and enterprise networks. Real-life example: A company firewall can block unwanted incoming network connections.

5.3 Types of Firewall

Definition: The main firewall types in this syllabus are Packet Filtering Firewall, Application-Level Gateway, and Circuit-Level Gateway. Each type controls communication at a different level.

3 Uses:

- Packet filtering checks network packets.
- Application-level gateways work with application traffic.
- Circuit-level gateways monitor communication connections.

Applications & Real-Life Example: Applications: Used in network security to control and filter communication. Real-life example: A packet filtering firewall can allow or block traffic based on information such as source or destination address.